AUTOMATED AI ENGINEERING GUIDE

n8n Multi-Source RSS AI Summarizer

Continuous Ingestion from BBC Tech, TechCrunch, The Verge & Wired into Google Sheets

ARCHITECTURE	N8N ENGINE	AI INFERENCE	STORAGE TARGET
Docker Containerized	v2.44.6 (Port 5678)	OpenRouter (Free Tier)	Google Sheets OAuth2

1. Multi-Source Pipeline Architecture

The workflow solves the single-feed limitation (where polling only BBC Tech repeatedly yielded the same 5 cached articles). It ingests 4 top-tier tech publications, balances 2 articles per publisher, produces structured 2-sentence summaries, and prevents duplicate records via URL-based upserting.



2. Workflows Included in Repository

Workflow Filename	Pipeline Type	Trigger	Deduplication	Target Columns
`workflows/rss-summarizer-multi.json`	**Multi-Source (4 Feeds)**	Manual Click	Yes (`appendOrUpdate` on Link)	Source, Title, Link, Published, Summary
`workflows/rss-summarizer-multi-scheduled.json`	**Multi-Source (4 Feeds)**	Schedule (Daily 08:00)	Yes (`appendOrUpdate` on Link)	Source, Title, Link, Published, Summary
`workflows/rss-summarizer.json`	Single-Source (BBC Tech)	Manual Click	Yes (`appendOrUpdate` on Link)	Source, Title, Link, Published, Summary
`workflows/rss-summarizer-scheduled.json`	Single-Source (BBC Tech)	Schedule (Daily 08:00)	Yes (`appendOrUpdate` on Link)	Source, Title, Link, Published, Summary

3. Docker Deployment Instructions

n8n runs containerized with persistent state in Docker volume `n8n-class_n8n_data`.

Docker Compose (Recommended)

```
# Start n8n in detached mode
docker compose up -d

# Verify container is healthy
docker compose ps

# Import all workflows into container
chmod +x ./scripts/import-workflow.sh
./scripts/import-workflow.sh
```

Persistent Storage Architecture

All n8n credentials, SQLite databases, and workflow definitions reside in the external Docker volume `n8n-class_n8n_data` mounted at `/home/node/.n8n`. Updating, stopping, or restarting the Docker image preserves all configurations.

4. Authentication & Credentials Setup

A. OpenRouter AI Setup

1. Sign in to openrouter.ai/keys and generate a free API key.
2. In n8n, navigate to **Credentials → Add Credential → OpenRouter API** and paste the key.
3. The workflow uses model `openrouter/free`, which routes to high-quality free community models.

B. Google Cloud OAuth2 Setup

To enable n8n to append and update records in Google Sheets:

1. **Enable Google APIs**: In Google Cloud Console, enable both **Google Sheets API** and **Google Drive API**.
2. **OAuth Consent Screen**:
   - User Type: **External**.
   - Under **Test users**, click **+ Add Users** and input your Google email. (Failing to do this triggers `Error 403: access_denied`).
3. **Create OAuth Credentials**:
   - Application type: **Web application**.
   - Authorized Redirect URI: `http://localhost:5678/rest/oauth2-credential/callback`.
4. **Google Sheet Destination**: Create a sheet titled `RSS Summaries` with row 1 headers: `Source | Title | Link | Published | Summary`.

5. Live Verification & Google Sheets Data

The multi-source workflow was executed end-to-end via headless container CLI (`docker exec -e N8N_RUNNERS_BROKER_PORT=5689 n8n n8n execute --id=shY8hRfrlFmK2Suq`). Verified sample rows logged in Google Sheets:

Source	Article Title	AI Generated Plain-English Summary
BBC Tech	UK to invest £100m in 'world first' AI Biosecurity Lab	The UK government plans to spend £100 million on a new laboratory that uses artificial intelligence to protect against biological threats.
TechCrunch	AI delivers ‘zero financial payoff’ to most companies	Despite massive investment and hype, artificial intelligence has provided no financial benefit to most US companies outside tech.
The Verge	Now you can watch Instagram Reels on your Apple TV	Instagram has quietly launched its dedicated TV app on Apple TV, bringing Reels and live streams directly to living room displays.
Wired	Scientists Trapped an Antimatter Particle for Months	Researchers successfully contained an antimatter particle for months, enabling breakthroughs in understanding why matter dominates the universe.

6. Production Troubleshooting Matrix

Symptom	Root Cause	Resolution
Single feed fetches same 5 items	Feed has not updated since last run; static Limit takes top 5.	Use `rss-summarizer-multi.json`. It checks 4 sources and deduplicates via `appendOrUpdate`.
Error 403: access_denied	Google OAuth in Testing mode without email in Test Users.	Add Google account in Google Cloud Console → OAuth consent screen → Test users.
Google Drive / Sheets API Disabled	APIs not activated in Google Cloud project.	Enable **Google Sheets API** and **Google Drive API** in Google Cloud Library.
columns.schema required	n8n v4.4+ requires column schema metadata in defineBelow mode.	All workflow files in `workflows/` include the full schema definitions matching the sheet.
OpenRouter 429 Rate Limit	High traffic on free model tier.	Workflow nodes include 3 automated retries with 2000 ms backoff.